Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/12/2015
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Universities Space Research Association		
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PI Address 2:	3600 Bay Area Blvd		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2004	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per K. George/HRP (Ed., 11/30/16) NOTE: End date changed to 10/31/2016 per K. George/HRP (Ed., 8/16/16) NOTE: Changed end date to 7/31/2016 as work is continuing per PI (Ed., 12/10/15) NOTE: Change in end date per PI (Ed., 3/12/2015)		
Key Personnel Changes/Previous PI:	September 2012: David Ware was removed from the protocol. October 2010: Buccello-Stout, Regina (Universities Space Research Association) removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Yarbough, Patrice (Universities Space Research Association) Zanello, Susana (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Flight Analogs Project (FAP) Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine, and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility, and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume, and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In 2014, Standard Measures were collected as part of the ongoing Campaign 11 (C11). A total of 8 studies participated in C11. These studies are listed below. • Physiological Factors Contributing to Post Flight Changes in Functional Performance, (J. Bloomberg, NASA) • Integrated Resistance and Aerobic Training Study (iRATS), (L. Ploutz-Snyder, Universities Space Research Association) • Testosterone supplementation as a countermeasure against musculoskeletal losses during space exploration, (R. Urban, University of Texas Medical Branch) • Effects of retronasal smelling, variety and choice on appetite & satiety (J. Hunter, Cornell University) • AD ASTRA: Automated Detection of Attitudes and States through Transaction Recordings Analysis, (C. Miller, Smart Information Flow Technologies) • Bed Rest as a Spaceflight Analog to Study Neuro-cognitive Changes: Extent, Longevity, and Neural Bases, (R. Seidler, University of Michigan) • Surveillance of Ocular Parameters and Visual Function in Bed Rest Subjects, (G. Vizzeri, University of Texas Medical Branch) • Integrated Resistance and Aerobic Exercise Training with Small Compact Exercise Equipment (Flywheel), (L. Ploutz-Snyder, Universities Space Research Association) For the Flywheel study only a subset of standard measures were collected to inform the investigator science. In Campaign 19 (C19), a single study was integrated with a minimal set of standard measures to inform the investigator science. This study is listed below. • Effects of long duration spaceflight on venous and arterial compliance in Astronauts, (S. Platts, NASA) All bed rest studies at the NASA Flight Analogs Research Unit at UTMB were completed on December 31, 2014. Data analysis continues and will be documented in subsequent reports.
Bibliography Type:	Description: (Last Updated: 05/16/2019)
Abstracts for Journals and Proceedings	Cromwell RL. "Countermeasures and Functional Testing in Head-Down Tilt Bed Rest (CFT 70)." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3108.pdf , Feb-2014
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Abstracts for Journals and Proceedings	Bloomberg JJ, Ballard KL, Batson CD, Buxton RE, Feiveson AH, Kofman IS, Lee SMC, Miller CA, Mulavara AP, Peters BT, Phillips T, Platts SH, Ploutz-Snyder LL, Reschke MF, Ryder JW, Stenger MB, Taylor LC, Wood SJ. "Body unloading associated with space flight and bed-rest impacts functional performance." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3163.pdf , Feb-2014
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Articles in Peer-reviewed Journals	Ploutz-Snyder LL, Downs M, Ryder J, Hackney K, Scott J, Buxton R, Goetchius E, Crowell B. "Integrated resistance and aerobic exercise protects fitness during bed rest." Med Sci Sports Exerc. 2014 Feb;46(2):358-68. http://dx.doi.org/10.1249/MSS.0b013e3182a62f85 ; PubMed PMID: 24441215 , Feb-2014
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